

AI WEATHERWISE API

PROJECT DOCUMENTATION TEMPLATE

Team ID

SWTID-2026-3698

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Project Name: AI Weatherwise API

1. INTRODUCTION

AI Weatherwise API is a smart weather information project designed to provide weather details in a simple and accessible way. The system uses weather API services to collect information for a selected location. The collected data can include temperature, humidity, wind speed, weather condition, pressure and forecast information.

The project combines API-based weather data with AI-oriented processing so that weather information can be presented in a more understandable form. Instead of depending on multiple websites or applications, users can use one system to obtain important weather information.

2. PROBLEM STATEMENT

Weather conditions can change frequently, and users need current information for travel, education, work, outdoor activities and daily planning. Traditional methods may require users to search through different sources. This can make the process time-consuming and difficult for users who want quick information.

The proposed AI Weatherwise API solves this problem by collecting weather information through an API and presenting the required details in a clear format. The system is intended to make weather checking faster, simpler and more user-friendly.

3. PROJECT OBJECTIVES

- To provide current weather information for a selected location.
- To retrieve weather data using an API.
- To display important weather parameters clearly.
- To provide simple and understandable weather information.
- To design a user-friendly weather application.
- To provide a foundation for future AI-based weather features.

4. EXISTING SYSTEM

In a general weather information system, users may depend on different weather websites, mobile applications or search engines. Information may be spread across different screens, and users may need to compare information manually. Such systems may not always provide a personalized or simplified explanation of the weather.

5. PROPOSED SYSTEM

The proposed AI Weatherwise API provides a centralized way to retrieve weather information. The user enters a city or location, the application sends a request to a weather API, receives the response, processes the required values and displays the result.

The system can be extended with AI-based features such as simple weather summaries, activity suggestions, alerts and natural-language explanations. These features can improve the usefulness of the application.

6. FUNCTIONAL REQUIREMENTS

- User should be able to enter a city or location.
- System should send a request to the weather API.
- System should receive weather information.
- System should process the API response.
- System should display temperature and weather condition.
- System should display humidity and wind information.
- System should show forecast information when supported.
- System should handle invalid or unavailable locations.

7. NON-FUNCTIONAL REQUIREMENTS

- Easy to use interface
- Fast response time
- Reliable API communication
- Secure handling of API keys
- Readable weather information
- Maintainable project structure
- Ability to support future improvements

8. TECHNOLOGIES USED

Frontend: HTML, CSS and JavaScript can be used to create the user interface.

Backend: A suitable backend/API integration layer can process requests and responses.

Weather API: A weather service provides real-time or forecast weather data.

AI: AI-based processing can be used to generate understandable weather summaries.

Development Tool: Visual Studio Code or another suitable development environment.

9. SYSTEM ARCHITECTURE

The system follows a simple request-and-response architecture. The user interacts with the application through the interface. The application sends the location request to the weather service. The weather API returns structured data. The application then extracts the required values and displays them to the user.

Layer	Responsibility
User Interface	Accepts location and displays results
Application/API Layer	Sends requests and processes responses
Weather Service	Provides weather information
AI Processing	Generates simple explanations or summaries
Output Layer	Shows useful weather details to the user

10. SYSTEM WORKFLOW

Step 1: User opens the application.

Step 2: User enters a city or location.

Step 3: Application creates an API request.

Step 4: Weather API receives the request.

Step 5: Weather API sends weather data.

Step 6: Application processes the response.

Step 7: AI layer can create a simple explanation.

Step 8: Final weather information is displayed.

11. PROJECT MODULES

11.1 User Interface Module

Provides a simple screen where users can enter a location and view weather information.

11.2 Location/Search Module

Accepts the user's city or location and prepares it for the API request.

11.3 Weather API Module

Connects the application with the weather service and retrieves the required weather data.

11.4 Data Processing Module

Extracts important values such as temperature, humidity, wind speed and weather condition from the API response.

11.5 AI Weather Explanation Module

Can convert technical weather data into a simple explanation that is easier for users to understand.

11.6 Error Handling Module

Handles invalid locations, missing information, API errors and network-related problems.

12. EXPECTED OUTPUT

For a selected location, the application is expected to show the location name, current temperature, weather condition, humidity, wind speed and other available weather information. The interface should present these values clearly so that users can understand the current conditions quickly.

13. SAMPLE TEST CASES

Test Case	Input	Expected Result
TC01	Valid city	Weather details displayed
TC02	Invalid city	Error message displayed
TC03	Empty input	Input validation shown
TC04	API unavailable	Suitable error message
TC05	Valid location	Temperature and condition shown

14. ADVANTAGES

- Simple and user-friendly.
- Provides weather information quickly.
- Uses API-based data retrieval.
- Reduces the need to check multiple sources.
- Can provide weather information in a structured format.
- Can be enhanced with AI features.
- Suitable as a foundation for a web or mobile application.

15. FUTURE ENHANCEMENTS

The project can be expanded with additional features such as severe weather alerts, voice-based weather assistance, multi-day forecasts, personalized activity suggestions, notification support, mobile application support, multiple language support and more advanced AI-based weather summaries.

16. SECURITY CONSIDERATIONS

API keys and other credentials should be stored securely and should not be exposed directly in public source code. The application should validate user input and handle API errors safely. Secure communication should be used wherever supported.

17. CONCLUSION

AI Weatherwise API is a useful project for retrieving and presenting weather information through an API. The system focuses on simplicity, quick access and understandable weather information. With future AI and notification features, the project can become a more complete weather assistance platform.

PROJECT TEAM

Role	Name
Team Leader	Pravena M
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